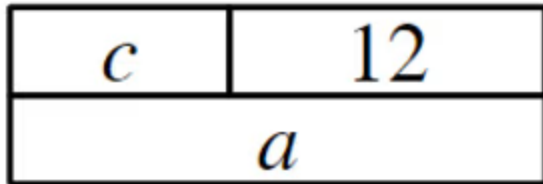




Equality in an equation

1 Which equations are represented by this bar model? Tick **2** correct answers



☐ $a + c = 12$

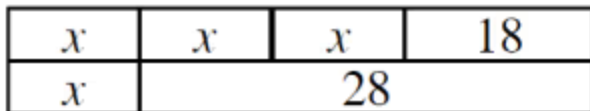
☐ $a - c = 12$

☐ $c - a = 12$

☐ $12 + a = c$

☐ $c = a - 12$

2 Which equations are represented by this bar model? Tick **2** correct answers



☐ $3x + 18 = 28$

☐ $3x + 18 - 28 = x$

☐ $x + 28 - 3x + 18 = 0$

☐ $x = x + 28 - 18 - 2x$

☐ $2x = 28 - 18 - x$

3 If $31 - x = 17$, which of these equations are also true? Tick **2** correct answers

☐ $31 - 17 = x$

☐ $17 - x = 31$

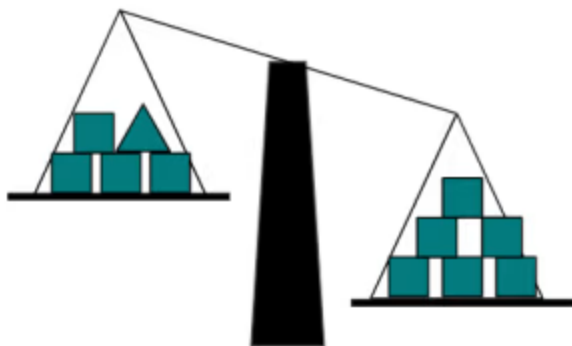
☐ $x + 17 = 31$

☐ $x = 31 + 17$

☐ $x = 17 - 31$



- 4 These scales are not balanced. The mass of all the square objects are the same. Which statements are definitely true? Tick **1** correct answer



- ☐ The triangle is lighter than 1 square.
- ☐ The triangle is lighter than 2 squares.
- ☐ The triangle has the same mass as 1 square.
- ☐ The triangle has the same mass as 2 squares.
- ☐ The triangle is heavier than 1 square.

- 5 The diagram shows a balanced scale on the left. When an operation is performed to the RHS, it is not balanced. What operation must be performed to the LHS to maintain equality? Tick **1** correct answer



- ☐ + 1
- ☐ + 3
- ☐ + x
- ☐ $\times 2$
- ☐ $\times 3$

- 6 The diagram shows a balanced scale on the left. When an operation is performed to the LHS, it is not balanced. What operation must be performed to the RHS to maintain equality? Tick **2** correct answers



- ☐ + 1
- ☐ - 1
- ☐ + x and - 1
- ☐ - 2x and - 1
- ☐ $\times \frac{1}{2}$

